

SHILPA VIJAY

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SUMMARY

Thermo-fluid engineer with expertise in designing custom heat transfer test systems, operating multi-sensor DAQ platforms, and validating CFD models against high-fidelity experimental data. Track record of delivering 200–400% heat transfer improvements through novel surface designs. Skilled in IR thermography, cold-wire anemometry, PIV, and multiphysics optimization for thermofluid applications. Experienced presenting technical findings to funding managers and at leading international venues. Comfortable navigating uncertainty and turning concepts into working solutions in fast-paced R&D environments.

EDUCATION

University of Southern California, Los Angeles, CA

Dec 2023

Ph.D. and M.S. Mechanical Engineering — *Focus: Heat Transfer, Fluid Dynamics*

College of Engineering Pune, India

May 2016

B.S. Civil Engineering

SKILLS

Thermal Characterization: IR thermography, cold-wire anemometry, thermocouples, pressure/flow sensors, DAQ systems, PIV, high-speed imaging, wind tunnel testing, uncertainty quantification

Modeling: MATLAB, Python, PyTorch, OpenFOAM, ANSYS Fluent, ICEM CFD, Bayesian optimization, Linux environment

Design: SolidWorks, OnShape, Autodesk Fusion, ANSYS Mechanical/Maxwell, additive manufacturing, rapid prototyping

EXPERIENCE

Postdoctoral Fellow | Center for Turbulence Research, Stanford University

Apr 2024 – Present

Selected for competitive CTR Postdoctoral Fellowship to lead independent thermal-fluid research.

- Designed and built modular heated-wall test section with 12+ embedded thermocouples, IR-transparent window, and cold-wire probe integration for boundary layer thermal measurements at Re up to 10,000.
- Developed calibration protocols for cold-wire anemometry (AN-1005), achieving $\pm 0.05^\circ\text{C}$ uncertainty; generated benchmark datasets validating CFD predictions within 8% error.
- Co-developed Design-by-Morphing framework for multiphysics optimization of porous surfaces, generating non-intuitive geometries from 4 baseline topologies using Bayesian optimization for thermofluid applications.
- Executed 500+ OpenFOAM CFD simulations to map permeability-geometry relationships; identified ~20% performance sensitivity to subtle topological features invisible to traditional parametric methods.
- Configured DAQ systems to synchronize thermal, pressure, and flow sensors at 10 kHz sampling rates for time-resolved turbulent heat flux measurements.
- Presented experimental validation strategies to Department of Energy/National Nuclear Security Administration program managers during site review for a \$25M+ center proposal.
- Participated in NASA Thermal and Fluids Analysis Workshop (TFAWS) at NASA Ames (2025), engaging with system-level thermal modeling and CFD validation practices.

PhD Researcher | University of Southern California

Aug 2016 – Dec 2023

Awarded 4-year Viterbi Graduate Fellowship (top incoming PhD students across engineering).

- Designed and fabricated custom partially-porous channel flow facility with integrated thermal management, pressure taps, and optical access for PIV diagnostics.
- Achieved 200–400% enhancement in convective heat transfer without proportional increase in pumping power through engineered porous surface architectures.
- Characterized anisotropic porous substrates using 3D-printed lattice structures; quantified permeability-porosity relationships across 15+ configurations.

- Validated experimental pressure drop measurements against ANSYS Fluent simulations; published results in *Experiments in Fluids* (2024), delivered 8 conference presentations at APS DFD and ASME.
- Finalist, USC Viterbi 3-Minute Thesis Competition (2023): Presented PhD research to non-specialist judges, selected across all engineering schools.

Materials Research Intern | Defense R&D Organization (DRDO), India

Aug 2015 – May 2016

- Optimized composite laminate thickness tapering using ANSYS Mechanical FEA, reducing material usage 15% while maintaining structural integrity.

Visiting Research Scholar | University of Southern California

May – Jul 2015

Awarded S.N. Bose Scholarship (<5% acceptance rate, top 55 students nationally in India).

- Executed 3-month experimental campaign characterizing hydraulic permeability of custom 3D-printed heterogeneous porous materials.

Attitude Controls Engineer | ISRO Student Satellite Team

May 2013 – May 2016

- Designed passive magnetic attitude control system components (magnetorquer rods, hysteresis dampers) for SWAYAM pico-satellite using ANSYS Maxwell.
- Presented Critical Design Review at ISRO Satellite Center; satellite successfully launched 2016. Team recognized by Prime Minister of India.

SELECTED PEER-REVIEWED PUBLICATIONS

- Vijay, S.,** Luhar, M. *Pressure drop measurements over anisotropic porous substrates in channel flow.* *Experiments in Fluids* 65, 135 (2024)
- Chavarin, A., Efstathiou, C., **Vijay, S.,** Luhar, M. *Resolvent-based design and experimental testing of porous materials for passive turbulence control.* *Int. J. Heat Fluid Flow* 86 (2020)
- Vijay, S.,** McKeon, B.J. *Experimental investigation of turbulent heat transfer over roughness elements.* 11th Int. Symp. Turbulence, Heat and Mass Transfer (THMT), Tokyo (2025)
- Vijay, S.,** McKeon, B.J. *Thermal footprint of boundary layer with a step change in roughness.* CTR Annual Research Briefs (2025)
- Lee, S., **Vijay, S.** *Topology-aware permeability modeling in structured porous media for passive flow control.* CTR Annual Research Briefs (2025)
- Vijay, S.,** McKeon, B.J. *Experimental investigation of heat transfer over roughness elements: design and preliminary results.* CTR Annual Research Briefs (2024)

PRESENTATIONS

- Invited:** "Towards designer porous surfaces for multi-functional passive flow control," UC Berkeley / Lawrence Berkeley Lab (Mar 2025)
- Invited:** "Flow and Thermal Transport at Porous Interfaces," CTR Tea Seminar, Stanford University (Apr 2024)
- "Thermal footprint of boundary layer under step change in surface roughness," 77th APS Division of Fluid Dynamics (Nov 2024)
- "Experimental investigation of heat transfer over roughness elements," Stanford Thermal & Fluid Sciences Affiliates Conference (Feb 2024)
- "Pressure Drop Measurements over Anisotropic Porous Substrates in Channel Flow," 76th APS Division of Fluid Dynamics (Nov 2023)
- "Tunable Porous Media for Multifunctional Passive Flow Control," Southern California Flow Physics Symposium, SDSU (Apr 2023)
- "Thermal Transport in Partially Porous Channel Flow," ASME Summer Heat Transfer Conference (2022)
- "Tunable Hydraulic and Thermal Properties via 3-D Printing," ASME Summer Heat Transfer Conference (2022)

SCIENCE COMMUNICATION

- Speaker, Pint of Science USA (2025):** Delivered public talk translating turbulence research for general audiences.
- Contributing Author, Stories of Women in Fluids (SOWIF):** Writing anthologies to inspire and mentor women and girls pursuing careers in fluid dynamics (2024–Present, books in press).